

AZ-104 Lab 02a – Manage Subscriptions and RBAC

Purpose of this Document

This document provides a complete, in-depth explanation of Azure governance and RBAC concepts covered in AZ-104 Lab 02a. It includes detailed explanations, real-world examples, and an interview-ready Q&A; section for quick revision.

Management Groups

Management groups provide a governance layer above Azure subscriptions. They allow administrators to organize subscriptions hierarchically and apply RBAC roles and Azure Policies at scale. Assignments at a management group scope are inherited by all child subscriptions.

- 1 Root management group is automatically created for every tenant.
- 2 Management groups simplify governance across multiple subscriptions.
- 3 RBAC and policies applied at management group scope are inherited.

Azure RBAC Fundamentals

Azure Role-Based Access Control (RBAC) is the authorization system used to control access to Azure resources. RBAC evaluates permissions based on security principals, role definitions, and scopes.

- 1 Security Principal: User, group, or service principal.
- 2 Role Definition: Collection of allowed and denied actions.
- 3 Scope: Management group, subscription, resource group, or resource.

Built-in vs Custom RBAC Roles

Built-in roles are predefined by Azure and cover common access scenarios. Custom roles are created when built-in roles provide more permissions than required, supporting the principle of least privilege.

- 1 Actions define allowed operations.
- 2 NotActions explicitly deny specific operations.
- 3 AssignableScopes define where the role can be assigned.

Role Assignment and Scope Inheritance

RBAC role assignments follow a hierarchical scope model. Assignments made at a higher scope are inherited by all lower scopes, reducing administrative overhead and ensuring consistent access control.

Monitoring with Activity Log

The Azure Activity Log records subscription-level events, including role assignments and custom role creation. It is a key tool for auditing and compliance.

Interview-Ready Questions and Answers

Q: What is Azure RBAC?

A: Azure RBAC is an authorization system that controls access to Azure resources by assigning roles to security principals at a specific scope.

Q: What are the components of RBAC?

A: RBAC consists of a security principal (user, group, or service principal), a role definition, and a scope where the role is applied.

Q: What is the difference between Azure RBAC roles and Microsoft Entra ID roles?

A: Azure RBAC roles control access to Azure resources at different scopes, while Microsoft Entra ID roles control access to directory and identity-related features at the tenant level.

Q: Why should roles be assigned to groups instead of users?

A: Assigning roles to groups simplifies access management, reduces administrative effort, and ensures consistent permissions as users join or leave the group.

Q: What are AssignableScopes in a custom RBAC role?

A: AssignableScopes define where a custom RBAC role can be assigned. Setting it at the management group scope allows the role to be inherited by all child subscriptions.

Q: Why are custom RBAC roles used?

A: Custom RBAC roles are used to enforce the principle of least privilege when built-in roles provide more permissions than required.

Q: How can RBAC changes be audited in Azure?

A: RBAC changes can be audited using the Azure Activity Log, which records role assignments and custom role creation events.

Q: What happens when a role is assigned at management group scope?

A: The role assignment is inherited by all subscriptions and resources under that management group.